PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE NAME	

CT GROUP	**1**	**4**				INDEX NUMBER	

CHEMISTRY **9647/03**

Paper 3 Free Response

22 September 2015

2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CT group on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.
A Data Booklet is provided.
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

Answer any **four** questions

1 Chlorine undergoes the following equilibrium in aqueous solution.

$$Cl_2(aq) + H_2O(l) \rightleftharpoons Cl^-(aq) + HOCl(aq) + H^+(aq) \quad \textit{equilibrium 1}$$

(a) **(i)** Write an expression for the equilibrium constant for this reaction, K_c, stating its units. [1]

(ii) When 33 g of Cl_2 was bubbled into 100 cm^3 of water, and allowed to reach equilibrium, the resulting $[Cl_2(aq)] = 4.57$ mol dm^{-3}.

Calculate the equilibrium values of $[Cl^-(aq)]$ and $[HOCl(aq)]$ in this solution and hence calculate a value for K_c. [2]

(iii) Describe and explain the effect of adding water to *equilibrium 1* above. [2]

(b) Chlorine is a strong oxidising agent which can oxidise thiosulfate ion readily to sulfate in the following overall equation.

$$4Cl_2 + S_2O_3^{2-} + 5H_2O \rightarrow 8Cl^- + 2SO_4^{2-} + 10H^+$$

The cell consisting of a $SO_4^{2-}(aq)$, $H^+(aq)$, $S_2O_3^{2-}(aq)|Pt(s)$ half-cell and a $Cl_2(g)|Cl^-(aq)|Pt(s)$ half-cell was set up, and shown using conventional cell notation below.

$$Pt(s)|S_2O_3^{2-}(aq), H^+(aq), SO_4^{2-}(aq) \,||\, Cl_2(g)|Cl^-(aq)|Pt(s)$$

(i) Construct an equation for the anode reaction. [1]

(ii) The cell is capable of producing an e.m.f of +1.04V. By using suitable data from the *Data Booklet*, calculate a value for the $E^\ominus$ of the $SO_4^{2-}/S_2O_3^{2-}$ electrode reaction. [2]

(iii) Explain how the addition of $Ba(NO_3)_2(aq)$ to the $SO_4^{2-}(aq)$, $H^+(aq)$, $S_2O_3^{2-}(aq)|Pt(s)$ half-cell, will affect the e.m.f. of the cell. [2]

(c) Compound **A**, $C_{11}H_{15}NO_2$, exhibits geometric isomerism.

It is insoluble in water but dissolves in hydrochloric acid to form a colourless solution. 1 mole of **A** reacts with Na to form 1 mole of H_2 gas. **A** produces a yellow precipitate when warmed with alkaline aqueous I_2. **A** decolourises aqueous Br_2, forming a white precipitate with the molecular formula of $C_{11}H_{15}Br_2NO_3$.

When **A** is treated with $K_2Cr_2O_7$ under suitable conditions, compound **B**, $C_{11}H_{11}NO_2$ is formed. Compound **B** forms a brick-red precipitate with Fehling's solution.

A reacts with hot aqueous $KMnO_4$ under suitable conditions to give **C**, $C_8H_7NO_4$, liberating a gas which gives a white precipitate with limewater.

Deduce the structures of compounds **A**, **B** and **C**, giving reasons for your answer.
[10]

[Total:20]

2 (a) Propanone reacts with iodine in the presence of an acid. The kinetics of the reaction

$$CH_3COCH_3 + I_2 \xrightarrow{H^+} ICH_2COCH_3 + HI$$

can be investigated experimentally by varying the concentration of the three substances involved and determining the time for the colour of the iodine to disappear. In this method, the rate at which the iodine concentration changes,

$$\text{rate of reaction} \propto \frac{\text{volume of aqueous iodine used}}{\text{time for colour of iodine to disappear}}$$

The following results were obtained in such an experiment.

experiment number	volume of 1 mol dm^{-3} propanone / cm^3	volume of 0.01 mol dm^{-3} iodine / cm^3	volume of 1 mol dm^{-3} sulfuric acid / cm^3	volume of water / cm^3	relative time for iodine colour to disappear
1	8	4	8	0	1
2	8	4	4	4	2
3	4	4	8	4	2
4	8	2	8	2	0.5

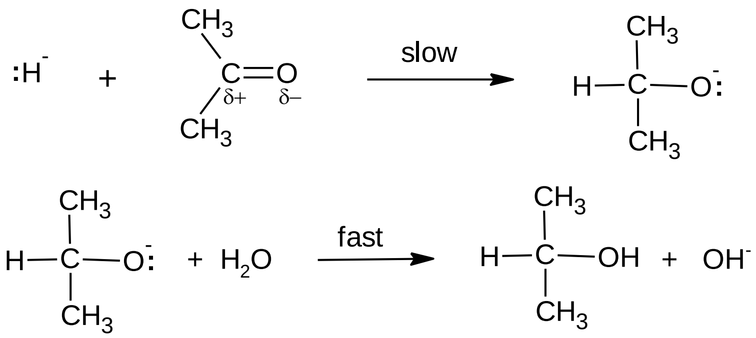
(i) Deduce the order of reaction with respect to propanone, iodine and hydrogen ions. [3]

(ii) Write the rate equation for the reaction. [1]

(iii) Sketch a graph to show the change in concentration of I_2 with time for experiments 1 and 3. Label each experiment clearly on your graph. [2]

(iv) The reaction between propanone and bromine proceeds by a similar mechanism. How would you expect the rate of this reaction to compare with that of the above reaction? Explain your answer. [1]

(b) Kinetics studies suggests that the mechanism for the reduction of propanone by lithium aluminium hydride involves the following two steps:



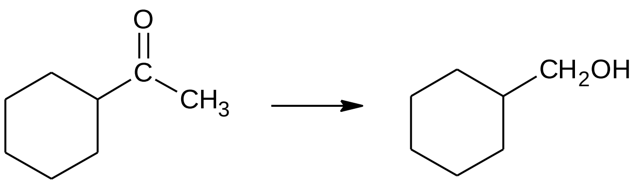
(i) Construct a rate equation for this reaction. [1]

(ii) Suggest how the rate of the reaction of CH_3CH_2CHO and $LiAlH_4$ compares with that of CH_3COCH_3, giving your reasoning. [2]

(iii) Draw the 'dot and cross' diagram, showing the electrons (outer shells only) in lithium aluminium hydride, $LiAlH_4$. In your diagram, use the symbol '□' for any additional electrons responsible for the overall negative charge of the anion. [2]

(iv) Suggest the shape of the anion in $LiAlH_4$. [1]

(c) In no more than 3 steps, outline how the following transformation can be achieved. State the reagents and conditions for each step, as well as the structures of any intermediates formed. [2]



(d) In the *aldol* reaction, a strong base (e.g. the **:**OH^- ion) is used to generate a nucleophile, $^-$**:**CH_2CHO, from ethanal as shown in the following equation.

$$CH_3CHO + OH^- \rightleftharpoons {}^-{:}CH_2CHO + H_2O$$

(i) Explain why the nucleophile $^-$**:**CH_2CH_3, cannot be generated from the reaction of ethane with hydroxide ion. [2]

(ii) By considering the nucleophile generated from ethanal and the equation for the *aldol* reaction below,

$$CH_3CHO + CH_3CHO \xrightarrow{OH^-} \underset{\text{(aldol)}}{CH_3CH(OH)CH_2CHO}$$

name and describe the mechanism for the aldol reaction. [3]

[Total:20]

3 (a) The water molecule can react in various ways: as an acid, as a base, as a nucleophile, as an oxidising reagent and as a reducing agent.

Study the following reactions and decide in which way water is reacting in each case. Explain your answers fully.

(i) $H_2O + NaH \rightarrow NaOH + H_2$ [1]

(ii) $2H_2O + 2F_2 \rightarrow O_2 + 4HF$ [1]

(iii) $2H_2O + C_2H_5CN + H^+ \rightarrow C_2H_5CO_2H + NH_4^+$ [1]

(iv) $2H_2O + 2Na \rightarrow 2NaOH + H_2$ [1]

(b) Acidified potassium manganate(VII), $KMnO_4$, is a strong oxidising agent which can oxidise both $\mathbf{X}^{2+}$ and $C_2O_4^{2-}$ in $\mathbf{X}C_2O_4$. One of the products of this reaction is CO_2.

37.50 cm^3 of 0.0200 mol dm^{-3} acidified $KMnO_4$ was found to completely oxidise 25.0 cm^3 of 0.050 mol dm^{-3} $\mathbf{X}C_2O_4$ solution.

(i) Write a balanced equation for the reaction between MnO_4^- and $C_2O_4^{2-}$. Hence, determine the volume of $KMnO_4$ needed to oxidise only $C_2O_4^{2-}$. [2]

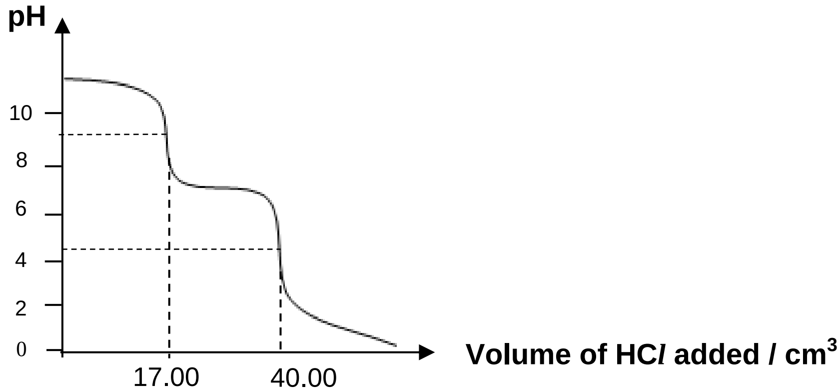
(ii) Using your answer in **(i)**, find the oxidation state of **X** in the product. [2]

(c) The CO_3^{2-} ion reacts with H^+ in two stages.

$$CO_3^{2-} \xrightarrow[pK_{b1}=3.67]{+H^+} HCO_3^{-} \xrightarrow[pK_{b2}=7.90]{+H^+} CO_2 + H_2O$$

A solution contains a mixture of aqueous sodium carbonate, Na_2CO_3 and sodium hydrogencarbonate, $NaHCO_3$.

25.0 cm^3 of the solution was titrated against 0.300 mol dm^{-3} hydrochloric acid, HC*l* using phenolphthalein as an indicator until a colour change was observed. The titration was then continued by using a second indicator, methyl orange. The titration curve for the reaction is given below.



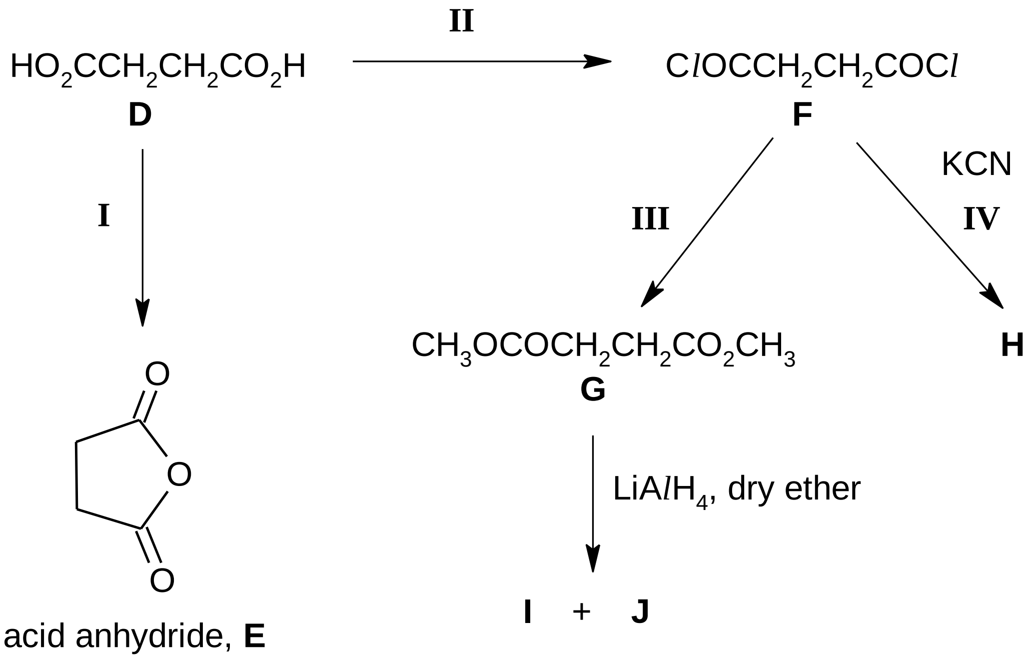
(i) State the colour changes observed at the first and second end-points. [1]

(ii) Calculate the concentration, in mol dm^{-3}, of Na_2CO_3 in the 25.0 cm^3 solution. [1]

(iii) From the graph, determine the volume of acid used to react with HCO_3^- originally present in the 25.0 cm^3 solution. Hence, determine the concentration of the HCO_3^- in the 25.0 cm^3 solution. [2]

(iv) Using your answer in **(ii)** and **(iii)** and any other information provided, calculate the pH of the original solution. [2]

(d) Consider the following reaction scheme involving butane-1,4-dioic acid.

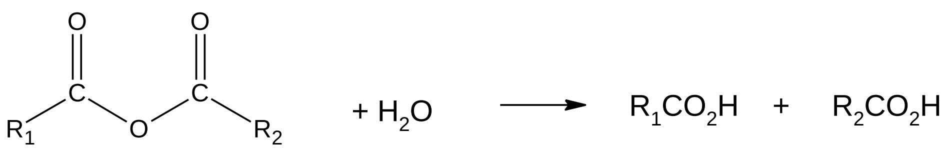


(i) Suggest the reagent and condition for step **II**. [1]

(ii) Draw the structures of the compounds, **H**, **I** and **J**. [3]

(iii) What type of reaction has occurred in the formation of the acid anhydride **E** in step **I**? [1]

(iv) Acid anhydrides undergo the same reactions as acyl chlorides, but at a slower rate. Whilst acyl chlorides react with water to produce a carboxylic acid and HC*l*, acid anhydrides react with water to produce two carboxylic acids as shown below.

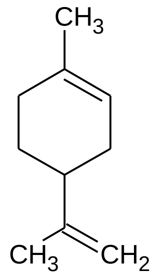


Suggest the structural formula of the product obtained when acid anhydride **E** reacts with ethanol instead of water. [1]

[Total: 20]

4 (a) Concentrated sulfuric acid reacts with solid potassium bromide to form two different bromine-containing products **K** and **L**.

Treating limonene, $C_{10}H_{16}$, separately with **K** and **L** produces compounds **M**, $C_{10}H_{18}Br_2$ and **N**, $C_{10}H_{16}Br_4$ respectively.



limonene

Separate reactions of compounds **M** and **N** with hot aqueous NaOH give **P**, $C_{10}H_{20}O_2$ and **Q**, $C_{10}H_{20}O_4$ respectively.

Q decolourises hot acidified purple $KMnO_4$, to yield **R**, $C_{10}H_{16}O_5$, but **P** does not.

(i) By identifying the bromine-containing products **K** and **L**, write equations for their formation. [2]

(ii) Write an equation for the reaction between **L** and hot aqueous NaOH. [1]

(iii) Draw the structural formulae of compounds **M, N, P, Q** and **R**. [5]

(iv) By quoting appropriate data from the *Data Booklet*, explain how the rate of reaction with aqueous NaOH will change when the bromine atoms of compound **M** are replaced with iodine atoms? [2]

(b) Describe what is observed when each of the halide ions, Cl^-, Br^- and I^- reacts with aqueous silver nitrate, followed by aqueous ammonia (dilute and concentrated). [5]

(c) Use the *Data Booklet* to describe the relative reactivity of the halogens as oxidising agents and relate this reactivity to the relevant $E^\ominus$ values. [2]

(d) The halogens form a number of interhalogen compounds, including the pale yellow liquid, BrF_3. This compound is a powerful fluorinating agent, converting silicon dioxide into silicon tetrafluoride and two other elements as products.

(i) Predict the shape of BrF_3. [1]

(ii) Suggest a balanced equation for the reaction. [1]

(iii) Use your equation to calculate the mass of BrF_3 needed to react with 10.0 g of silicon dioxide. [1]

[Total: 20]

5 (a) (i) The melting point of SO_2 is -72 ^{o}C while that of Al_2O_3 is 2072 ^{o}C. Account for the difference in their melting points, in terms of their structure and bonding. [3]

(ii) MgO, Al_2O_3 and SiO_2 exist as white solids. If a sample of one of the oxides was provided as a white powder, describe the reactions you could carry out on the powder to determine which of the three oxides it was. [3]

(b) (i) When solid selenium dioxide undergoes sublimation, an equilibrium exists between the two phases.

$$SeO_2(s) \rightleftharpoons SeO_2(g)$$

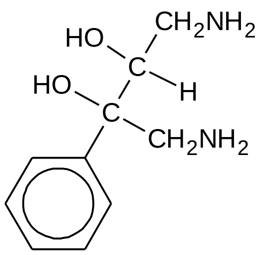
The standard enthalpy change, $\Delta H^{\ominus}$, and the standard entropy change, $\Delta S^{\ominus}$, of sublimation are +111 kJ mol^{-1} and +183 J K^{-1} mol^{-1} respectively. Predict whether the sublimation of solid selenium dioxide is spontaneous at room temperature. [2]

(ii) In 1932, Riley and co-workers published their findings on the use of SeO_2 to introduce carbonyl functional group into ketones. The following are some examples of Riley oxidation.

$$\underset{\text{propanone}}{CH_3COCH_3} \xrightarrow{SeO_2} \underset{\text{2-oxo-propanal}}{CH_3COCHO}$$

$$\underset{\text{butanone}}{CH_3COCH_2CH_3} \xrightarrow{SeO_2} \underset{\text{butan-2,3-dione}}{CH_3COCOCH_3}$$

Starting from phenylethanone, devise a three-step synthesis for compound **S**. You are to use Riley oxidation in one of your steps. For each step, give the reagents and conditions and give the structures of the intermediates formed.



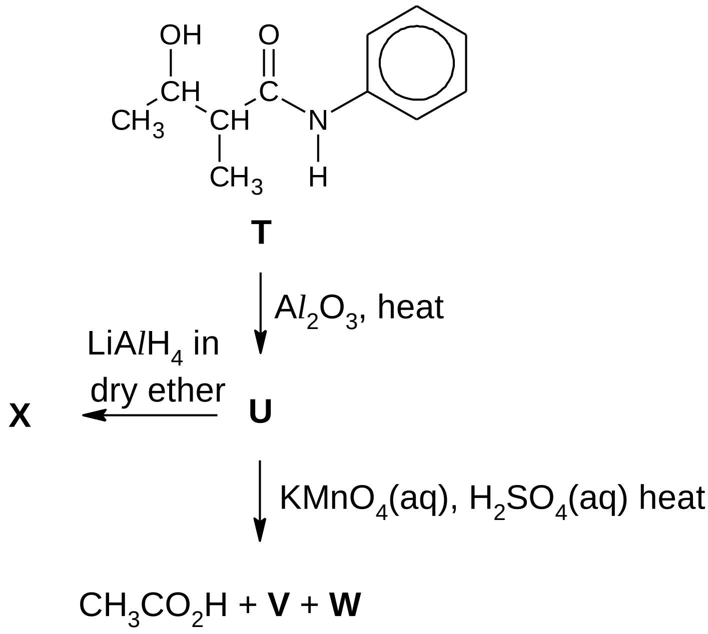
compound **S**

[3]

(c) Compound **T** undergoes dehydration on heating with Al_2O_3 to form compound **U**.

U undergoes reaction with hot acidified $KMnO_4$, to form three products, **V**, **W** and ethanoic acid. **V** was found to give a yellow precipitate with alkaline aqueous iodine.

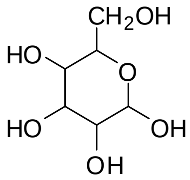
On treatment with $LiAlH_4$, **U** forms **X**.



(i) Give the structures of **U, V, W** and **X.** [4]

(ii) What type of stereoisomerism can **U** exhibit? Draw the displayed formulae of the stereoisomers of **U**. [2]

(d) Glucose, an important simple sugar or monosaccharide, is required by the cells in our body as a source of energy and metabolic intermediate.



Glucose, $C_6H_{12}O_6$, undergoes complete combustion to form carbon dioxide and water.

(i) Using the molecular formula of glucose, write a balanced equation for the complete combustion of glucose. [1]

(ii) Use bond energy values from the *Data Booklet* to calculate the enthalpy change of combustion of glucose.[Rather than the C=O value given in the *Data Booklet*, use a value of 805 kJ mol^{-1} for the bond energy of each C=O bond in CO_2.] [2]

[Total:20]

End of Paper